

MECHANICAL TECHNOLOGY

WELDING

LEARNER WORKBOOK

YEAR 2/2024

TERM 1

NAME OF SCHOOL

:KWA-THEMA SKILLS SCHOOL

NAME OF LEARNER

:.....

CLASS

:.....

TEACHER

: MR W. CHANGOROMA



WORKSHEET 1

NAME OF LEARNER: DATE:

Safety Sections (3 Marks)

QUESTION 1

TRUE OR FALSE QUESTIONS

Employers should ensure tools used by workers are		True	False
a)	Suitable for the work area where the task is to be performed		
b)	Maintained in good working condition		
c)	Suitable for the tasks undertaken		

QUESTION2

Layout Sections (9 Marks)

Acetylene, anchored, curtains and screens, tip-over, welding flash arc, safe, ventilated, airflow

- a) Welding booths prevent surrounding workers from being exposed to (1)
- b) If your workshop do not have welding booths, you use and.....to protect other workers from welding flash arc. (2)
- c) LPG lasts longer than and its cylinders don't need to be replaced as often. (1)
- d) Cylinder gases are stored in a location and secured to prevent(2)
- e) Cylinder is held securely in place by chain. (1)
- f) Ensure welding work area is appropriately..... through natural or a fixed/portable extraction system. (2)

QUESTION 3

Personal safety (8 Marks)



	Identity PPE according to number	State its purpose
1		
2		
3		
4		
5		
6		
7		

8. What does PPE stands for?

A1	SHAPE & COLOUR	WRITE THE ME	20 Marks	SIGN (2 Marks each)	WRITE THE NUMBER OF THE SYMBOL THAT
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			MATCHES THE MEANING				
PROHIBITION							
MANDATORY							
WARNING							
SAFE							
You must do, Where to go in an emergency, Do not do, Risk of danger, Carry out the action given by the sign, You must not, Caution, Hazard ahead, Stop, The safe way							
 Hand protection must be worn	1	 No smoking	2	 First aid	3	 Risk of explosion	4

WORKSHEET 2

NAME OF LEARNER: DATE:

QUESTION 1

SIGNS AND SYMBOLS (12 Marks)

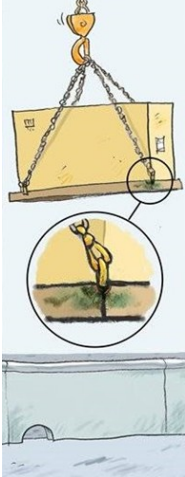
WORKSHEET 3

NAME OF LEARNER: DATE:

QUESTION 2

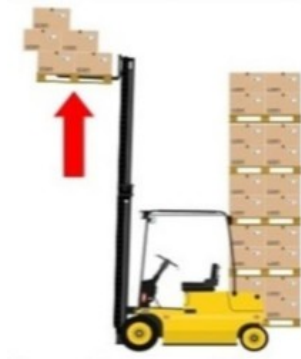
SIGNS AND SYMBOLS (12 Marks)

UNSAFE ACTS & CONDITIONS

Rusting lifting gear/chain		
Indicate below the pictures weather unsafe act or unsafe condition		
		

QUESTION 3.1

Indicate what the picture shows under it unsafe act, unsafe condition, accident or near miss



Question 3.2

What are Unsafe Conditions?

	Indicate by ticking true or false	true	false
	Unsafe conditions refer to actions or behaviors that deviate from established safety protocols.		
	Unsafe acts encompass physical elements within the workplace that have the potential to cause harm		
Examples of unsafe conditions are:-			
	poor housekeeping		
	Neglecting to wear safety goggles, helmets, gloves, or other necessary protective gear.		
	malfunctioning equipment		
	Using machinery or equipment without adequate knowledge		
	Ignoring established safety protocols, such as lockout/tagout procedures or confined space entry requirements.		
	inadequate lighting or ventilation,		
	Engaging in horseplay or reckless behaviour		

	improper handling of hazardous materials.		
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What is an unsafe act?

WORKSHEET 4

NAME OF LEARNER: DATE:

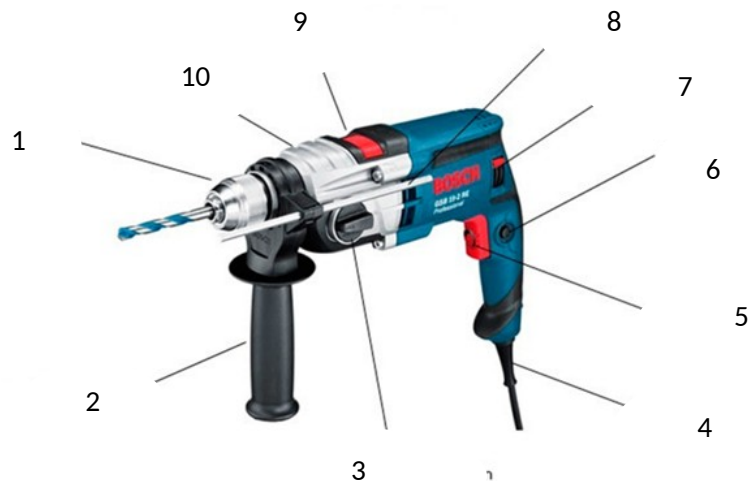
	Choose the area where the PPE protects on the body of the worker	Function
1		Safety helmet to protect against falling objects such as wooden beams and loose bricks etc.
2		Safety glasses to protect against dusts, mists, fumes and gases that may be harmful to eyes or cause irritation.
3		Earmuff and earplug used to protect the wearer's ears from excessive noise pollution, dust or temperature variations.
4		Safety shoes to prevent injuries of feet from falling objects such as bricks etc.
5		Safety gloves to protect hands from cuts and abrasions which might result from contact with sharp objects.
6		This is used for protecting entire body from heat, cold or other harmful effects of working environment such as electrical shock by wearing appropriate clothing (e.g., rubber boots).
7		Safety masks to protect against the inhalation of harmful dusts, mists, fumes and gases.
Respiratory protection // Eye and face protection // Foot protection // Body protection // Head protection // Hand protection // Hearing protection		

WORKSHEET 5

NAME OF LEARNER: DATE:

QUESTION 1

POWER TOOLS (10 Marks)



PART NO.	PART NAME ACCORDING TO THE NUMBER
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
Depth stop // strain relief cord // trigger lock // variable speed trigger // reverse function // chuck // clutch // drill or hammer function //side handle // safety clutch	

QUESTION 2

POWER TOOLS (10 Marks)



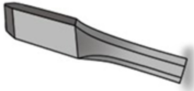
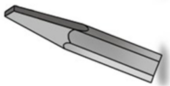
Part No.	Part name
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
Safety mask// air outlet // external carbon brush // inlet // metal head // cutting disc // power cable protection // rear switch // spindle lock // removable vice handle	

WORKSHEET 6

NAME OF LEARNER: DATE:

YEAR: 2/2024

Hand tools (10 Marks)

No.	Hand Tool	Name
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		